Prior Sensitivity in Bayesian Logistic Regression: A Small-Sample Medical Study

William Chen
University of California, Irvine
Instructor: Prof. Weining Shen
Stat 205P: Bayesian Data Analysis

Abstract

In Bayesian logistic regression, the choice of prior is inconsequential when data are plentiful but decisive when they are scarce. We demonstrate this using the `birthwt` dataset ($n = 189$), comparing three priors (diffuse $\mathcal{N}(0, 100)$, weakly informative $\mathcal{N}(0, 2.5)$, and a clinically informative prior) across the full sample and subsamples of $n \in \{20, 40, 80, 189\}$. At full data all three are interchangeable (AUC ≈ 0.747; LOOIC difference < 4). Under small samples they are not: at $n = 20$ the diffuse prior collapses under complete separation ($\hat{\beta}_{\texttt{smoke1}} = 203$) while the regularizing priors stay stable. Crucially, how quickly a coefficient escapes prior sensitivity is governed not by sample size but by the number of positive cases for that predictor: smoking stabilizes by $n = 40$, whereas hypertension (only 6.3% of patients) remains unstable until $n \approx 80$. For small clinical datasets the recommendation is concrete: prefer a weakly informative prior to a diffuse one, and treat rare predictors with particular caution, since they are the last to shed the prior's influence.

1 Introduction

In Bayesian analysis the prior's influence grows as data shrink: when samples are scarce it can substantially shape, and even distort, the posterior—a concern that is especially acute in clinical research. This report presents a prior sensitivity analysis of Bayesian logistic regression using the `birthwt` dataset ($n = 189$), comparing three priors (diffuse $\mathcal{N}(0, 100)$, weakly informative $\mathcal{N}(0, 2.5)$, and a clinically informative prior) refitted across $n \in \{20, 40, 80, 189\}$. We find that sensitivity is governed not by sample size alone but by predictor rarity, with direct implications for clinical datasets where the most important risk factors are often uncommon.

2 Data

2.1 Dataset Overview

The `birthwt` dataset ($n = 189$) was collected at Baystate Medical Center, Springfield, MA (1986). The binary outcome is low birth weight (< 2.5 kg), with 130 normal and 59 low birth weight cases (31.2% positive rate). All analyses use `stan_glm` (`rstanarm`), fit with 4 chains $\times$ 1000 iterations.

2.2 Predictors

Predictors include maternal age, pre-pregnancy weight, race, smoking status, prior preterm labor, hypertension history (`ht`), uterine irritability (`ui`), and first-trimester physician visits. Among the binary predictors, positive cases are sparse and uneven: 74/189 for smoking, 28 for uterine irritability, and only 12 for hypertension—a spread that becomes consequential under small samples.

3 Methods

3.1 Prior Specification

Three prior configurations were evaluated:

Model	Coefficients	Intercept
Diffuse	$\mathcal{N}(0,\ 100)$	$\mathcal{N}(0,\ 100)$
Weak	$\mathcal{N}(0,\ 2.5)$	$\mathcal{N}(0,\ 10)$
Informative	`smoke`, `ht`: $\mathcal{N}(1,\ 0.5)$; others $\mathcal{N}(0,\ 1)$	$\mathcal{N}(0,\ 5)$

The informative prior encodes clinical knowledge: $\mathcal{N}(1, 0.5)$ for `smoke` and `ht` corresponds to OR ≈ 2.7 (± 2 SD: OR $\in [1.4, 7.4]$), consistent with published estimates [1, 2]. All other predictors use $\mathcal{N}(0, 1)$.

3.2 Analyses

We conduct three analyses: (1) full-data prior comparison via posterior coefficients, LOO-CV, and AUC; (2) a subsampling experiment refitting all models at $n \in \{20, 40, 80, 189\}$ to track convergence across sample sizes, including a 100-draw repeated subsampling robustness check; and (3) predictive performance assessed via LOOIC and Pareto $\hat{k}$ diagnostics. Convergence was verified via $\hat{R} \leq 1.002$ and trace plots.

4 Results

4.1 Full Data ($n = 189$): Priors Have Little Effect

At $n = 189$, all three models produce nearly identical posterior estimates and converge well. Table 1 reports posterior means for the two clinically relevant predictors, and Figure 1 shows the full coefficient comparison across all predictors.

	Diffuse		Weak		Informative	
Variable	Mean	95% CI	Mean	95% CI	Mean	95% CI
`smoke1`	0.983	[0.21, 1.83]	0.933	[0.13, 1.74]	0.915	[0.31, 1.52]
`ht1`	1.994	[0.63, 3.50]	1.827	[0.49, 3.30]	1.311	[0.51, 2.13]

Table 1: Posterior means and 95% credible intervals (log-odds) for the two clinically targeted predictors at $n = 189$.

The informative prior modestly shrinks `ht1` (from 1.994 to 1.311), but this difference is small in practical terms. Table 2 reports LOO and AUC for all three models; the differences are negligible, confirming that prior choice has little practical effect at $n = 189$.

Model	LOOIC	AUC
Diffuse	223.85	0.746
Weak	223.32	0.747
Informative	220.43	0.748

Table 2: LOO model comparison and AUC at full data ($n = 189$). Lower LOOIC indicates better fit.

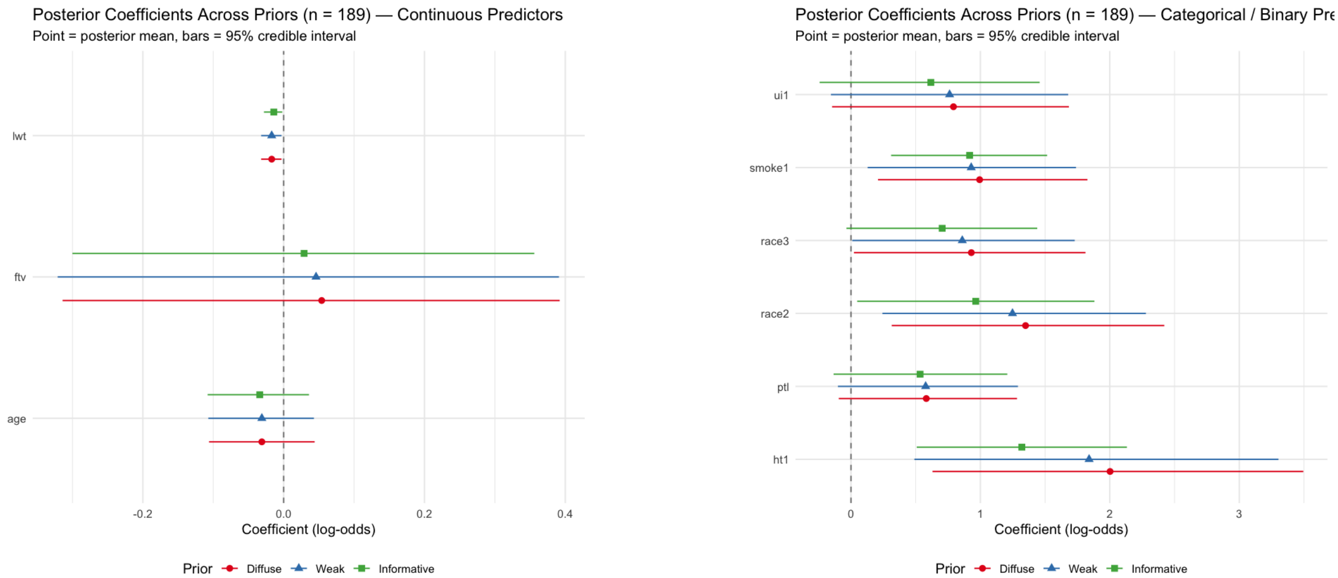

Figure 1: Posterior coefficients across priors at $n = 189$. Left: continuous predictors. Right: categorical/binary predictors. Points = posterior mean; bars = 95% credible interval.

4.2 Subsampling: Prior Sensitivity at Small n

To examine how prior influence changes with sample size, we refit all three models at $n \in \{20, 40, 80, 189\}$. We focus on `smoke1` and `ht1`, the two predictors for which the informative prior encodes directional knowledge, as these show the largest divergence across priors at small n (Figure 2).

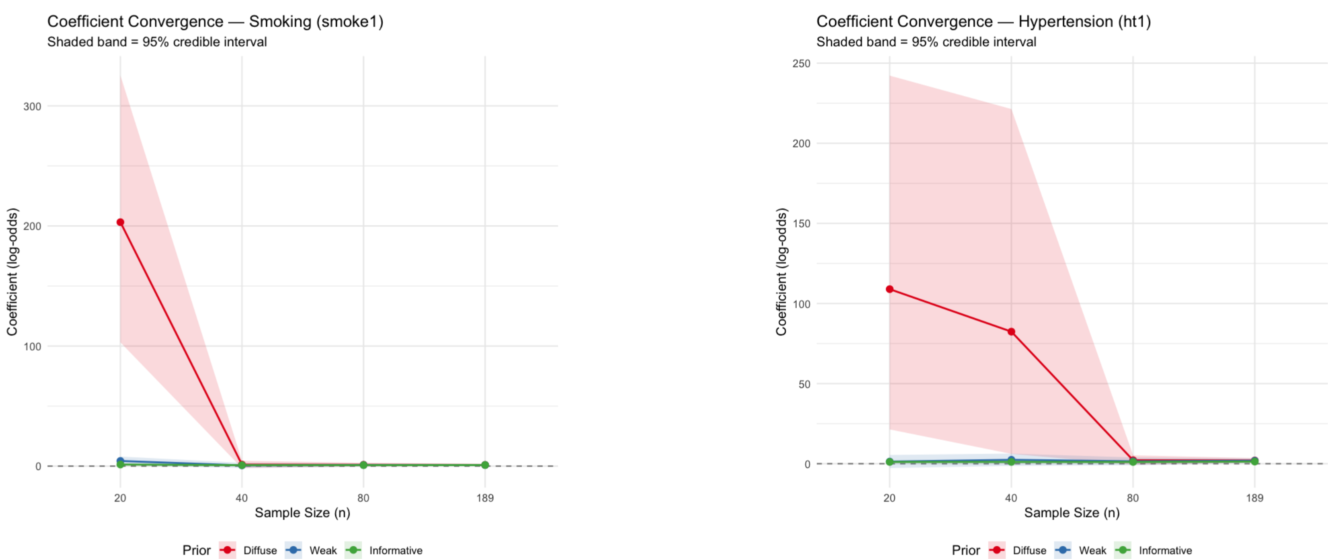

Figure 2: Posterior mean convergence. Left: `smoke1`. Right: `ht1`. Shaded bands = 95% credible intervals.

Under the diffuse prior, complete separation produces implausible estimates at $n = 20$ ($\hat{\beta}_{\texttt{smoke1}} = 203$, $\hat{\beta}_{\texttt{ht1}} = 109$). `smoke1` recovers by $n = 40$, but `ht1` remains unstable ($\hat{\beta} = 82.4$): hypertension affects only 12 of 189 patients (6.3%), so small subsamples rarely contain enough $\texttt{ht} = 1$ cases to resolve separation. By $n = 80$, all models stabilize. The weak and informative priors remain in clinically plausible ranges throughout. (These coefficient values are from a single representative draw with `set.seed(42)`; the robustness check below confirms they are typical rather than exceptional.)

Robustness Check: Repeated Subsampling

To confirm that these thresholds are not artefacts of a single random draw, we repeated the subsampling 100 times per n with independent seeds and recorded the proportion of runs in which the diffuse prior produced $|\hat{\beta}| > 10$ (Figure 3). At $n = 20$, `ht1` explodes in 31.1% of draws (and is undefined—no `ht`$= 1$ case sampled—in a further 26%). `smoke1` and `ui1` largely stabilize by $n = 40$ (explosion rates $\leq 3\%$), while `ht1` persists at 17.6% at $n = 40$ and reaches zero only at $n = 80$. The single-draw results above are therefore representative, not coincidental.

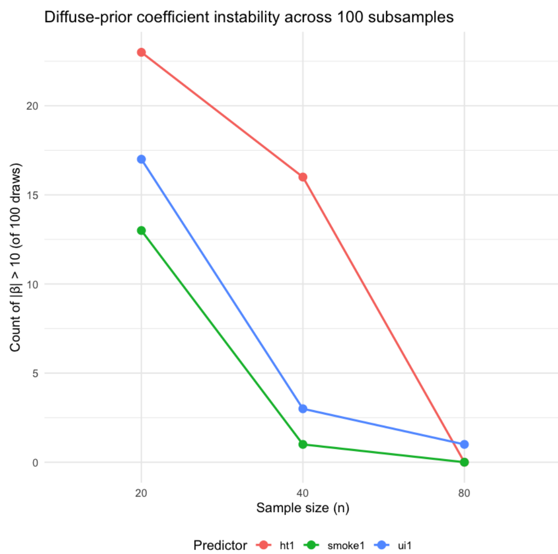

Figure 3: Subsamples (out of 100) where diffuse prior yields $|\hat{\beta}| > 10$. `ht1` is last to stabilize.

4.3 Predictive Performance

In-sample AUC is unreliable under separation; we use LOOIC and Pareto $\hat{k}$ diagnostics instead (Table 3). At $n = 20$ the diffuse prior's AUC $= 1.0$ reflects overfitting, confirmed by 20 problematic $\hat{k}$ values and a spuriously low LOOIC of 8.3. The informative prior remains reliable throughout (1 problematic observation at $n = 20$) and achieves the lowest LOOIC at every sample size, including full data (Informative $= 220.4$, Weak $= 223.3$, Diffuse $= 223.8$).

n	Model	AUC	LOOIC	$\hat{k} > 0.7$
20	Diffuse	1.000	8.3	20
20	Weak	0.952	28.1	5
20	Informative	0.905	27.1	1
40	Diffuse	0.808	65.3	4
40	Weak	0.799	61.1	0
40	Informative	0.750	56.3	1
80	Diffuse	0.789	105.9	0
80	Weak	0.785	102.4	1
80	Informative	0.777	98.2	0

Table 3: In-sample AUC, LOOIC, and Pareto $\hat{k} > 0.7$ count by sample size and prior. At $n = 20$ the diffuse prior's LOOIC and AUC are both unreliable.

5 Discussion

Two distinct mechanisms. Prior sensitivity at small n operates through two channels. The first is *separation stabilization*: when no positive cases are sampled, the likelihood is uninformative and a diffuse prior lets coefficients diverge—a problem frequentist MLE shares. A weakly informative prior acts as a regularizer, keeping estimates in a plausible range until more data arrive. The second is *gentle posterior shrinkage*: even without separation, an informative prior pulls estimates toward clinical expectations, as seen in `ht1` shifting from 1.994 (diffuse) to 1.311 (informative) at $n = 189$. These are related but distinct, and both are present in this analysis.

Limitations. Results derive from a single dataset; generalizability requires validation. At very small n, LOO is unreliable (Pareto $\hat{k} > 0.7$); k-fold cross-validation is more appropriate in such settings.

6 Conclusion

Prior choice matters most when data are scarcest. At $n = 20$ a diffuse prior produces unstable, uninterpretable posteriors while a weakly informative prior restores stability; by $n = 189$ the differences are negligible. Crucially, the threshold is not uniform: at $n = 20$ no predictor is reliable; by $n = 40$, two of three have stabilized—only hypertension remains, and it persists until $n = 80$, confirmed across 100 repeated draws. The governing factor is not sample size but the number of positive cases: smoking (74) and uterine irritability (28) recover together, while hypertension (12) is last.
For small clinical datasets the takeaway is direct: make a weakly informative prior the default, and treat rare predictors with particular caution—they are the last to shed the prior's influence.

References

[1] Simpson, W.J. (1957). A preliminary report on cigarette smoking and the incidence of prematurity. *American Journal of Obstetrics and Gynecology*, 73(4), 807–815.

[2] Shah, N.R., & Bracken, M.B. (2000). A systematic review and meta-analysis of prospective studies on the association between maternal cigarette smoking and preterm delivery. *American Journal of Obstetrics and Gynecology*, 182(2), 465–472.